CP255 - W8_S2 Line Chart Best Practices Answer Key

Your Turn:
Dataset: Weekly average commute time (minutes) recorded at 8 measurement points.
Steps:

- **Run the Before cell**
- Add dots to the line using marker='o' and the other marker arguments
- **Bonus:** try marker='s' instead of 'o' — what shape do you get?

Answer:

Python

```
# Your Turn - replace ... with your answers

fig, ax = plt.subplots(figsize=(9, 4))

ax.plot(
  weeks,
  commute,
  color='#e67e22',
  linewidth=2,
  marker='s',        # circle marker
  markersize=7,     # try 7 or 8
  markerfacecolor='white',  # try 'white' for open circles
  markeredgewidth=2  # try 2
)

ax.set_ylabel('Avg Commute Time (minutes)')
ax.set_title('Weekly Average Commute Time')
ax.spines[['top', 'right']].set_visible(False)
plt.tight_layout()
plt.show()
```

Your Turn:
Dataset: Annual average rent index for a city, 2005–2023.
Steps:

- **Run the Before cell** — notice how the crowded labels look
- Replace all labels with just three: 2005, 2020 (labeled '2020\nCOVID-19'), and 2023

- Add a dashed vertical reference line at 2020

Answer:

```python
# Your Turn — reduce x-axis labels and add a reference line

fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(rent_years, rent_index, color='#8e44ad', linewidth=2)

# Replace all labels with just three key moments
ax.set_xticks([2005, 2020, 2023])
ax.set_xticklabels(['2005', "2020\n COVID-19", "2023"])  # use \n for a two-line label

# Add a dashed vertical line at 2020
ax.axvline(
  x=2020,
  color='gray',
  linewidth=1,
  linestyle='--'
  )


ax.set_ylabel('Rent Index (2005 = 100)')
ax.set_title('Annual Average Rent Index, 2005–2023')
ax.set_xlim(rent_years[0], rent_years[-1])
ax.spines[['top', 'right']].set_visible(False)
plt.tight_layout()
plt.show()
```

Your Turn:
Dataset: Annual cycling infrastructure miles built in three cities over 12 years.
Steps:

- **Run the Before cell**
- Remove ax.legend()
- After the plot loop, add ax.text() for each line at its endpoint
- Match each label color to its line color
- Extend ax.set_xlim() to make room for the labels

Answer:

```python
# Your Turn — remove legend, add direct labels

fig, ax = plt.subplots(figsize=(8, 4))

for (city, values), color in zip(city_data.items(), city_colors):
  ax.plot(bike_years, values, color=color, linewidth=2)

  # Add label at the end of each line
  ax.text(
    bike_years[-1] + 0.25,      # x: just past the last year
    values[-1],      # y: last value of this line
    city,      # label: city name
    color=color,  # match the line color
    va='center',
    fontsize=9,
    fontweight='bold'
  )


ax.set_xlim(bike_years[0], bike_years[-1]+1)  # extend right edge to fit labels
ax.set_ylabel('Bike Lane Miles Built')
ax.set_title('Annual Cycling Infrastructure by City, 2012–2023')
ax.spines[['top', 'right']].set_visible(False)
plt.tight_layout()
plt.show()
```

Dataset: Annual park visitation (thousands) across six city parks over 15 years.
The story: Riverside Park has seen a dramatic decline.
Steps:

- **Run the Before cell**
- Set FOCUS = 'Riverside Park'
- In the loop, draw the focus line in a strong color and all others in gray
- Add a direct label only to the focus line
- Add a subtitle to the title that states the finding

Answer:

```python
# Your Turn: highlight the focus park, mute the rest

FOCUS = 'Riverside Park'

fig, ax = plt.subplots(figsize=(11, 5))

for park, values in parks.items():
  if park == FOCUS:
    # Focus: strong color, thicker line, higher zorder
    ax.plot(park_years, values, color='#e74c3c', linewidth=2.5, zorder=5)
    # Direct label at the end of the line
    ax.text(park_years[-1] + 0.5, values[-1], park, color="#e74c3c", va='center', fontsize=9,
fontweight='bold')
  else:
    # Background: gray, thin, slightly transparent
    ax.plot(park_years, values, color='#cccccc', linewidth=1, alpha=0.6, zorder=1)

ax.set_xlim(park_years[0], park_years[-1] + 2)  # extend for label
ax.set_ylabel('Annual Visitors (thousands)')
ax.set_title('Annual Park Visitation, 2008–2023\nRiverside Park attendance has significantly
decreased')  # add your subtitle with \n
ax.spines[['top', 'right']].set_visible(False)
plt.tight_layout()
plt.show()
```

Your Turn:
Dataset: Annual pedestrian injury rate for 12 city districts over 18 years.
Steps:

- **Run the Before cell to see the spaghetti chart**
- Build a small multiples chart using plt.subplots(nrows=3, ncols=4, sharey=True)
- Use .flatten() and enumerate() to loop through panels
- Add a gray band for the years 2020–2021
- Add a figure title using fig.suptitle()

Answer:

Python

```python
# Your Turn: replace ... with your code and convert to small multiples

fig, axes = plt.subplots(
  nrows=3,
  ncols=4,
  figsize=(14, 9),
  sharey=True
  )

axes_flat = axes.flatten()

for i, (district, values) in enumerate(ped_districts.items()):
  ax = axes_flat[i]

  # Gray band for COVID years
  ax.axvspan(2020, 2021, color="#f2f2f2", zorder=0)

  ax.plot(
    ped_years,
    values,
    color='#2980b9',
    linewidth=1.5
    )

  ax.set_xlim(
    ped_years[0],
    ped_years[-1]
    )

  ax.set_title(
    district,
    fontsize=9,
    fontweight="bold"
    )

  ax.spines[["top", "right"]].set_visible(False)

fig.suptitle(
  "Pedestrian Injury Rate by District, 2005–2023",
  fontsize=13,
```

```
    fontweight="bold",
    y=1.01
)
plt.tight_layout()
plt.show()
```